

A INDUSTRIAL TRAINING PROJECT REPORT

on

“full stack web-development”

From

“Kota Smart Citizen”

Duration: from 28 June to 10 August

Submitted in partial Fullfillment of the requirement for the

Award of the diploma of engineering

(computer science & engineering)

To



BOARD OF TECHNICAL EDUCATION, RAJASTHAN



GOVERNMENT POLYTECHNIC COLLEGE

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

DCM road, Kota(raj) – 324007

Academic year: 2025-26

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CANDIDATE DECLARATION

“As 3rd (CSE) year students, we take full ownership of the project

“Online medical Appoientment booking website” and its contents.

Submitted to **THE DEPARTMENT OF THE COMPUTER SCIENCE**

GOVT. POLYTECHNIC COLLEGE, KOTA (RAJASTHAN).

For DIPLOMA IN COMPUTER SCIENCE.IN

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Submission date: 23 September 2025

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Kota Smart Citizen

A unit of SSPR INC.

Internship Certificate

This certificate is awarded to

Mr./Miss. BHUMIKA RATHOR D/o Mr. MUKESH RATHOR

**In recognition of successful completion of an internship in
Web Development and Computer Language Concepts**

At

Kota Smart Citizen

During the period of

28 JUNE 2025 to 10 AUGUST 2025

This internship provided valuable experience in:

- Skills developed during the internship.
 - (Front-end development using HTML, CSS, and JavaScript)
 - (Back-end development concepts like databases and server-side scripting)
 - (Problem-solving and analytical skills)
 - (Understanding of web development best practices)
 - (Teamwork and communication skills)

We are confident that this experience will be a valuable asset to future endeavors.

(Director)

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Abstract

The rapid growth of technology has transformed the way healthcare services are delivered. Traditional methods of booking medical appointments often involve long waiting times, manual record-keeping, and lack of accessibility. To overcome these challenges, this project focuses on the development of a **Medical Appointment Booking System** that allows patients and doctors to connect through a dynamic and user-friendly web application.

The system is developed using **Laravel (PHP framework)** as the backend and **HTML, CSS, JavaScript, and Bootstrap** for the frontend. It provides separate role-based dashboards for patients and doctors, ensuring that both user groups can manage their respective activities efficiently. Patients can register, update their profiles, search doctors by category, view doctor details, book appointments in available time slots, and provide reviews and ratings. Doctors, on the other hand, can manage their profiles, consultation modes, availability slots, and monitor patient feedback.

The application ensures secure authentication and database management using **MySQL** integrated with Laravel's Eloquent ORM. By digitizing the appointment process, the system reduces manual workload, saves time for both patients and doctors, and enhances the overall healthcare experience.

This project not only demonstrates the integration of frontend and backend technologies but also showcases how web applications can solve real-world problems in the medical field.

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1. Introduction

Healthcare is one of the most essential sectors where timely access to doctors and medical facilities plays a critical role in saving lives and improving patient outcomes. Traditionally, patients have had to rely on manual processes such as visiting hospitals physically or making phone calls to book an appointment. These methods are not only time-consuming but also prone to errors, mismanagement, and inconvenience, especially in urban areas where both patients and doctors face heavy workloads.

With the advancement of digital technologies, web-based medical appointment systems have emerged as an effective solution to bridge the gap between doctors and patients. Such systems provide a structured platform where patients can easily search for doctors, check their availability, and book appointments at their convenience. At the same time, doctors can manage their schedules, consultation modes, fees, and view patient reviews in an organized manner.

The project “**Medical Appointment Booking Website**” has been developed with the objective of making the appointment process simple, reliable, and efficient. This system enables role-based access where patients and doctors have personalized dashboards. Patients can register, update profiles, book appointments, and rate doctors, while doctors can set their availability, manage their professional details, and monitor patient feedback.

The application has been designed using a **Laravel (PHP framework) backend** integrated with **MySQL database** for dynamic data handling, and a **responsive frontend** developed with HTML, CSS, JavaScript, and Bootstrap. This combination ensures a smooth user experience along with secure authentication and efficient data management.

In short, the project aims to bring healthcare services closer to people by minimizing waiting time, reducing manual efforts, and making healthcare more accessible through an interactive web application.

2.Literature Review / Existing System

- ❖ In the existing healthcare appointment process, most patients still rely on manual or semi-digital methods to connect with doctors. The most common practices include:
 - Walk-in Appointments:

Patients directly visit hospitals or clinics to book appointments. This method often results in long waiting times, crowding at reception desks, and lack of transparency in scheduling.

- **Telephonic Booking:**

Some hospitals provide appointment booking through phone calls. Although it reduces physical effort, it is still dependent on the availability of hospital staff, prone to miscommunication, and does not guarantee real-time slot confirmation.

- **Basic Online Systems:**

A few healthcare providers have websites or mobile applications, but many of them are limited to static information such as hospital address, contact details, and doctor names. They do not provide dynamic features like live slot booking, patient reviews, or role-based dashboards.

- ❖ **Limitations of Existing Systems**

- Lack of centralized platform for doctors and patients.
- No real-time tracking of doctor availability.
- Absence of role-based dashboards and profile management.
- Limited accessibility and poor user interface in traditional systems.
- Dependency on manual staff for confirmation and scheduling.

- ❖ **Proposed System Advantage**

The proposed **Medical Appointment Booking Website** eliminates these issues by providing:

- A dynamic role-based platform for both patients and doctors.
- Online registration and secure login system.
- Real-time slot booking and management.
- Review and rating features for better patient feedback.
- Centralized database for storing and managing medical appointment details.
- User-friendly responsive interface accessible across devices.

This system not only improves convenience for patients but also reduces administrative workload for doctors and healthcare providers, making the appointment process more efficient and reliable.

3. Project Overview

The **Medical Appointment Booking Website** has been developed as a complete web-based solution to simplify and digitalize the process of scheduling medical appointments. The primary aim of the project is to create an interactive platform where both patients and doctors can connect seamlessly, reducing manual efforts and saving time for both parties.

The system follows a **role-based structure**, where patients and doctors have different access rights and functionalities:

❖ For Patients:

- Register and login to the system.
- Manage their profile details and upload profile images.
- Search for doctors by category, specialization, or name.
- View doctor details such as availability, consultation fees, hospital address, and social media links.
- Book appointments in available time slots.
- Rate and review doctors based on their experience.

❖ For Doctors:

- Register and login with role-specific access.
- Manage personal details, specialization, consultation modes, and fees.
- Set available time slots for patients to book.
- View patient reviews and feedback for service improvement.
- Doctor can also explore and connect with other doctors.

❖ General Features:

- A **Home Page** with navigation bar, categories, image slider, “Meet Our Doctors” section, and footer.
- A **dynamic doctor listing page** where users can explore doctors by specialization.
- A **Doctor detail page** with contact number, hospital details, map integration, and social links.
- Every user have personalized dashboards.
- Secure authentication and personalized dashboards for patients and doctors.
- A responsive and user-friendly interface to ensure accessibility across devices.

Technology Stack:

- **Frontend:** HTML, CSS, JavaScript, Bootstrap
- **Backend:** Laravel (PHP Framework)
- **Database:** MySQL with Eloquent ORM
- **Development Tools:** VS Code, Composer, XAMPP

This project not only provides a smooth booking experience for patients but also helps doctors in efficiently managing their appointments and professional presence. The overall workflow ensures transparency, reliability, and convenience in the healthcare appointment process.

4. System Requirements

To successfully design, develop, and deploy the **Medical Appointment Booking Website**, certain hardware and software resources are required. These requirements ensure smooth functioning of both the development environment and the deployed application.

4.1 Hardware Requirements

- ❖ The hardware requirements for development and testing are as follows:
 - **Processor:** Intel Core i3 / AMD Ryzen 3 or above
 - **RAM:** Minimum 4 GB (8 GB recommended for smoother performance)
 - **Hard Disk:** Minimum 250 GB (SSD preferred for faster processing)
 - **Monitor:** Standard 15-inch or larger display
 - **Internet Connection:** Stable broadband connection for package installation, updates, and testing

4.2 Software Requirements

- ❖ The project requires the following software tools and technologies:
 - **Operating System:** Windows 10 / Linux Ubuntu / macOS (any modern OS)
 - **Backend Framework:** Laravel (PHP Framework)
 - **Programming Language:** PHP (with Composer dependency manager)
 - **Frontend Technologies:** HTML5, CSS3, JavaScript, Bootstrap
 - **Database:** MySQL (XAMPP or MySQL Workbench)
 - **Code Editor/IDE:** Visual Studio Code (VS Code)
 - **Version Control:** Git / GitHub for project versioning and collaboration
 - **Web Server:** Apache (via XAMPP) or Laravel built-in server

- **Browser:** Google Chrome / Mozilla Firefox for testing and debugging

4.3 Additional Tools

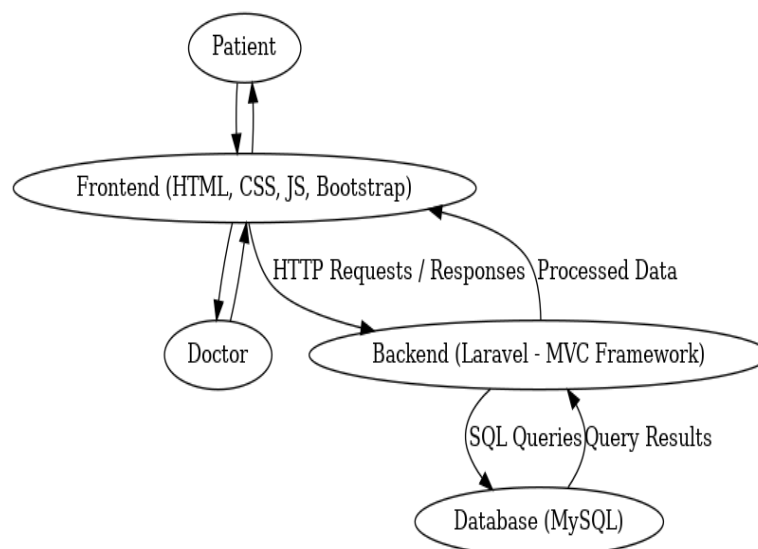
- **Postman:** For API testing (if required in future enhancements)
- **Draw.io / Lucidchart:** For preparing ER diagrams and flowcharts
- **Canva / Figma:** For UI design and prototyping

5. System Design

- ❖ System design defines the overall architecture, workflow, and data structure of the project. It ensures that both the frontend and backend components work together smoothly to achieve the objectives of the **Medical Appointment Booking Website**. The design phase translates the functional requirements into technical blueprints using diagrams and structured planning.

5.1 System Architecture

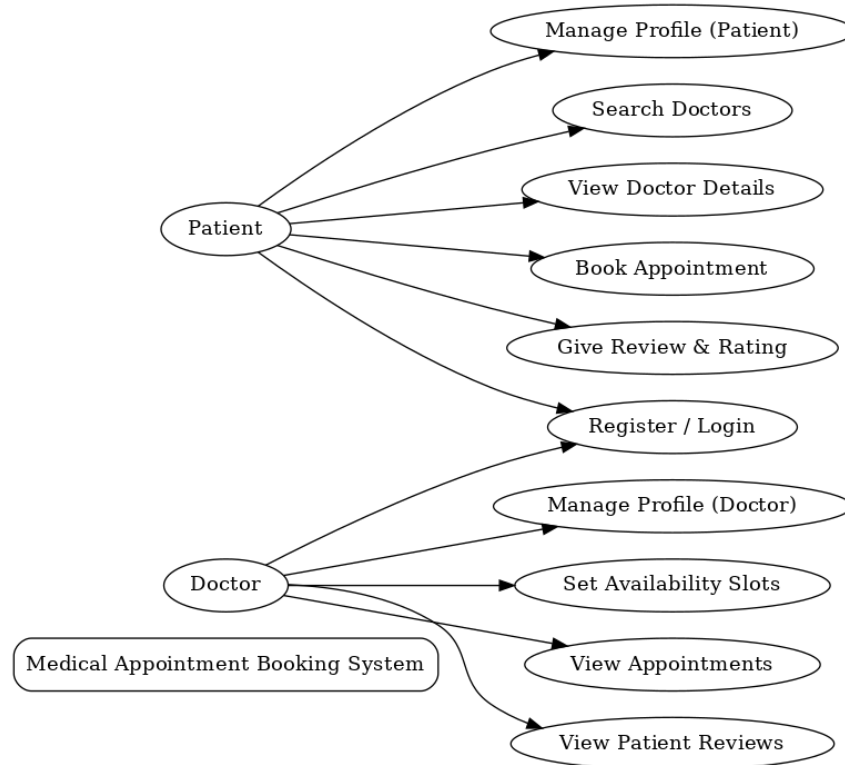
- ❖ The project is developed on the **Model-View-Controller (MVC)** architecture of the Laravel framework:
 - **Model:** Handles database interaction through Laravel's Eloquent ORM.
 - **View:** Manages user interface and presentation (HTML, CSS,).
 - **Controller:** Connects Models and Views, processes requests, and executes business logic.
- ❖ This layered approach improves **scalability, code reusability, and maintainability**.



System Architecture Diagram

5.2 Use Case Diagram

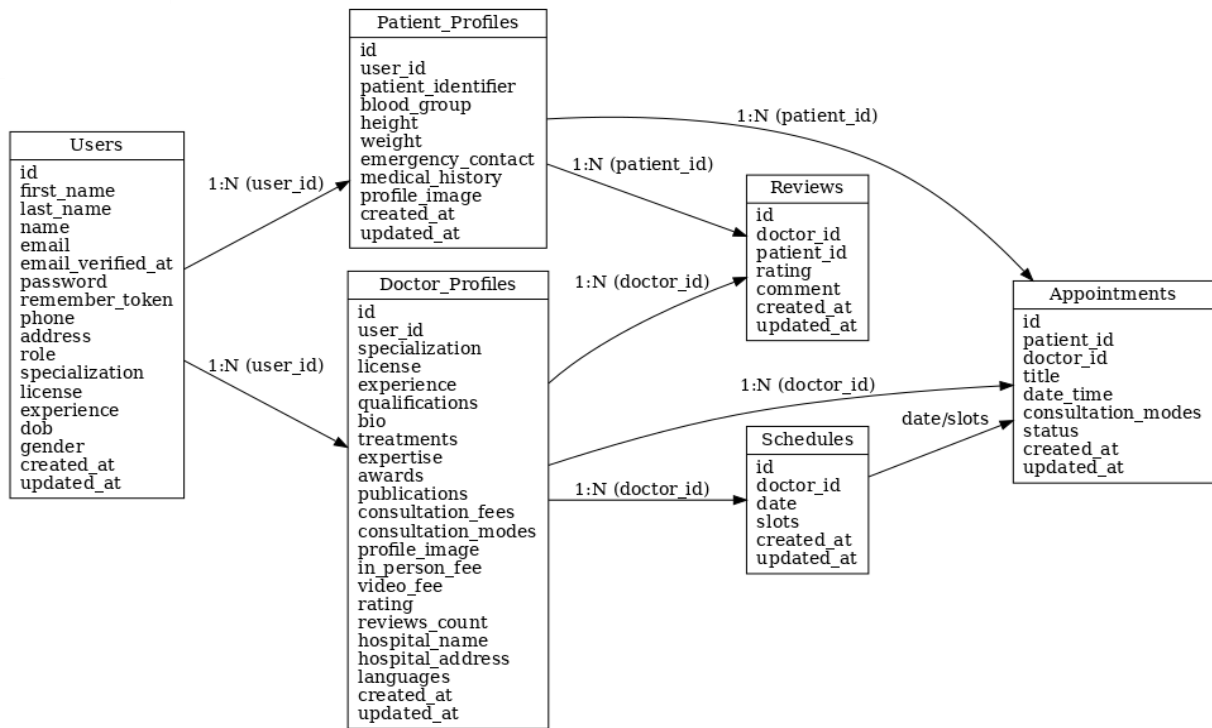
- ❖ The system supports two primary actors: **Patients** and **Doctors** (Admin can also be extended if required).
- **Patients can:** Register, login, manage profile, search doctors, view doctor details, book appointments, and give reviews.
- **Doctors can:** Register, login, manage profile, set available slots, view appointments, and check patient reviews.



Use case diagram

5.3 Entity Relationship (ER) Diagram

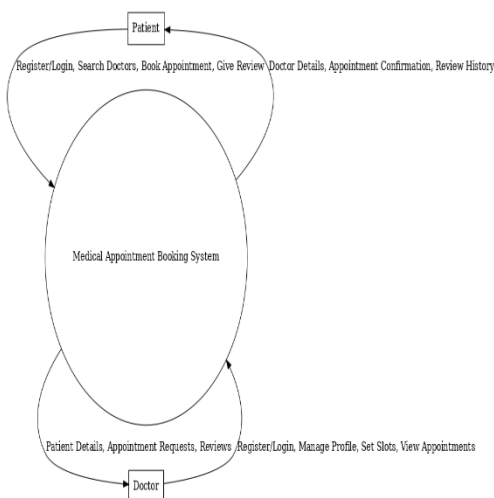
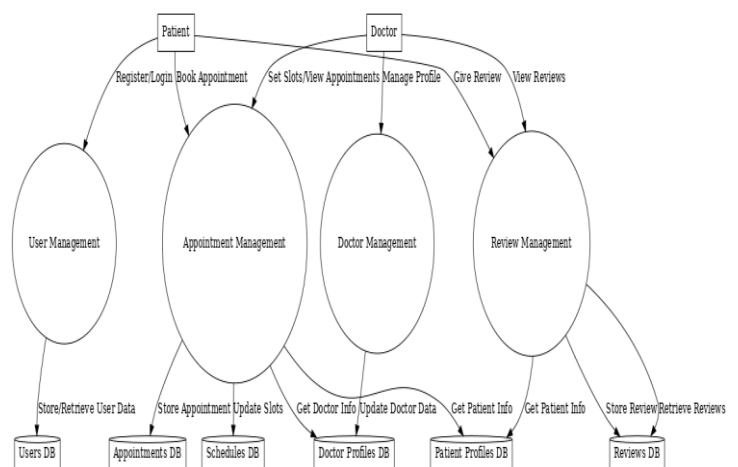
- ❖ The ER diagram defines the database structure. Major entities include:
 - **Users Table** – Stores login credentials and role (patient/doctor).
 - **Doctors_profiles Table** – Stores doctor details (specialization, hospital, consultation fee, availability).
 - **Patients_profiles Table** – Stores patient details and profiles.
 - **Appointments Table** – Stores appointment booking details (date, time, status).
 - **Reviews Table** – Stores patient feedback and ratings.
 - **Schedule table**- store time slot for specific date for doctor schedule.

**ER Diagram**

5.4 Data Flow Diagram (DFD)

❖ The Data Flow Diagram explains how data moves within the system:

- **Level 0 (Context Diagram):** User interacts with system → System communicates with database.
- **Level 1:** Patient registers/login → system validates credentials → dashboard opens → booking process connects with doctor availability slots.

**level 0****level 1**

5.5 Database Schema

- ❖ The database schema of the system is designed using relational tables to manage users, doctors, patients, appointments, reviews, and schedules efficiently.
 - **Users** – Stores general details of both doctors and patients such as name, email, phone, role, specialization, license, experience, etc.
 - **Patient_Profiles** – Contains medical information of patients like blood group, height, weight, medical history, and emergency contact.
 - **Doctor_Profiles** – Holds professional details of doctors including qualifications, bio, consultation fees, hospital details, and ratings.
 - **Appointments** – Manages booking information between patients and doctors with date, time, consultation mode, and status.
 - **Reviews** – Stores feedback from patients about doctors with ratings and comments.
 - **Schedules** – Maintains doctors' availability by date and time slots.
- ❖ **Relationships:**
 - Each **User** is either a Patient or a Doctor.
 - One **Doctor** can have many Appointments, Reviews, and Schedules.
 - One **Patient** can book multiple Appointments and give Reviews.

6. Module Description

- ❖ The Medical Appointment Booking System is divided into several functional modules. Each module handles specific operations and interacts with other modules to provide a seamless experience. The major modules of the system are described below:

6.1 User Module

- ❖ This module manages all the basic user-related operations.
 - New users can **register** as patients or doctors.
 - Users can **log in** to access their respective dashboards.
 - Role-based authentication ensures that doctors and patients have different privileges.
 - Users can **update profile information** including personal details and profile image.

6.2 Patient Module

- ❖ This module is dedicated to patients and provides the following features:
 - Manage patient profile including medical history, blood group, and emergency contact.
 - Search for doctors by **specialization, category, or name**.
 - View doctor details including hospital name, consultation fees, contact information, and available slots.
 - Book, cancel, or reschedule appointments.
 - Provide **reviews and ratings** for doctors after consultation.

6.3 Doctor Module

- ❖ The doctor module allows doctors to manage their professional information and interact with patients.
 - Maintain doctor profile including specialization, qualifications, experience, hospital details, and consultation fees.
 - Set availability slots using the **Schedule Module**.
 - View patient appointments and update status (pending, confirmed, completed).
 - Access reviews and ratings provided by patients to improve services.

6.4 Appointment Module

- ❖ This module handles the appointment booking process.
 - Patients can select doctors and book appointments based on available slots.
 - Each appointment stores details like date, time, consultation mode (online/offline), and status.
 - Doctors can confirm or reject appointment requests.
 - Appointment history is maintained for both patients and doctors.

6.5 Review Module

- ❖ The review module ensures feedback and quality monitoring.
 - Patients can give ratings and comments to doctors after appointments.
 - Reviews help other patients choose suitable doctors.
 - Doctors can view patient reviews for performance improvement.

6.6 Schedule Module

- ❖ This module manages doctors' availability.
 - Doctors can set their available **dates and time slots**.
 - Patients can only book appointments based on these available slots.
 - Prevents double booking and ensures proper time management for doctors.

6.7 Admin Module

- ❖ If an admin panel is implemented, the admin can:
 - Manage users and doctors.
 - Approve or deactivate doctor profiles.
 - Monitor appointments and system activities.

7. System Implementation

- ❖ System implementation is the process of converting the system design into a working application. In this project, the medical appointment booking system was implemented step by step, starting from environment setup, database creation, frontend development, backend coding, and finally integration and deployment.

7.1 Development Environment Setup

- **Framework:** Laravel (PHP framework) was used for backend implementation.
- **Server:** XAMPP/Laragon was configured for running Apache and MySQL.
- **Database:** MySQL was selected as the relational database for storing data.
- **IDE/Editor:** Visual Studio Code was used for writing and debugging the code.
- **Version control:** Git was used to maintain project versions.

This environment provided a stable base for efficient coding and debugging.

7.2 Frontend Implementation

- ❖ The frontend is the interface that patients and doctors interact with. It was implemented using **HTML, CSS, and JavaScript**.
 - **Home Page:** Includes navigation bar, categories section, doctor slider, about section, and footer.
 - **Authentication Pages:** Login and Registration pages were designed for both patients and doctors.

- **Doctor Listing Page:** Shows doctors by category or specialization, with a “View Details” option.
- **Doctor Detail Page:** Displays doctor’s profile, consultation modes, fees, contact details, hospital location, and reviews.
- **Appointment Booking Page:** Patients can view available slots and book appointments dynamically.
- **Responsive Design:** CSS and JavaScript were used to make the system mobile-friendly.

7.3 Backend Implementation

- ❖ The backend was developed using **Laravel Framework**, which follows **MVC (Model-View-Controller)** architecture.
 - **Models:** Represent database tables such as Users, Doctor Profiles, Patient Profiles, Appointments, Reviews, and Schedules.
 - **Controllers:** Handle user requests and return the correct response. For example, AppointmentController manages appointment booking logic.
 - **Routes:** Define URL structures and connect them to controllers. Example: /appointments/book maps to the booking function.
 - **Middleware:** Used for authentication and role-based access. Patients, doctors, and admin have different privileges.
 - **Blade Templates:** Laravel’s Blade engine was used to integrate backend data with frontend pages dynamically.

7.4 Role-Based Dashboard

- **Patient Dashboard:** Allows patients to manage their profile, view past and upcoming appointments, cancel bookings, and provide reviews.
- **Doctor Dashboard:** Enables doctors to update their profile, set consultation slots, manage patient appointments, and check received reviews.
- **Admin (optional):** Can manage users, doctors, and overall system settings.

7.5 Database Connectivity

- The application uses **MySQL** as the database.
- **Eloquent ORM** in Laravel simplifies database operations.

- Relationships such as **One-to-Many** (Doctor → Appointments), **Many-to-One** (Appointment → Patient), and **One-to-One** (User → Doctor Profile/Patient Profile) were implemented.
- Database migrations were used for table creation and schema updates.

7.6 Integration

- ❖ Frontend and backend were integrated to make the system fully dynamic.
 - Forms from the frontend (login, register, appointment booking) send requests to Laravel controllers.
 - Controllers process the data and update the database.
 - Data is retrieved and displayed dynamically (e.g., available doctors, patient appointments, doctor reviews).
 - Proper validation and error handling ensure secure and smooth integration.

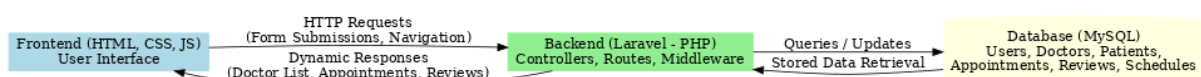
7.7 Security Features

- Passwords are stored using **bcrypt hashing** for security.
- **CSRF tokens** are used in Laravel forms to prevent cross-site request forgery.
- **Authentication Guards** ensure only logged-in users can access role-specific dashboards.
- Input validation prevents invalid data entry and SQL injection.

7.8 Deployment

- ❖ The system can be deployed on a local server (XAMPP/Laragon) or hosted on an online platform. Deployment involves:
 1. Uploading Laravel project files to the server.
 2. Importing the database schema.
 3. Configuring .env file for database and server settings.
 4. Running migrations and seeding initial data.

This completes the development cycle and makes the system ready for end-users.



System Implementation Flow

8. Testing

- ❖ Testing is a crucial step in software development to ensure that the system works as expected and meets the user requirements. The medical appointment booking system was tested at different levels to check functionality, performance, and reliability. Both manual and automated testing approaches were considered wherever applicable.

8.1 Unit Testing

- Each module of the system (User, Patient, Doctor, Appointment, Review, Schedule) was tested individually.
- Example: Testing the registration form to ensure that new users can register only with valid inputs (unique email, strong password).
- Functions like login, appointment booking, and review submission were tested separately.

8.2 Integration Testing

- After unit testing, modules were integrated together and tested.
- Example: When a patient books an appointment, the system updates the appointment table and notifies the doctor's dashboard.
- Ensured smooth data flow between frontend, backend, and database.

8.3 Functional Testing

- Focused on testing whether the system meets all the defined requirements.
- Verified features like:
 - Patient registration and login
 - Doctor profile management
 - Appointment booking and cancellation
 - Review submission and display
- All functional requirements performed correctly.

8.4 Role-Based Testing

- Different roles (Patient, Doctor, Admin) were tested to ensure proper access control.
- Patients could not access doctor features, and doctors could not use patient functionalities.

- Only authenticated users were allowed to book appointments and give reviews.

8.5 User Interface Testing

- The frontend was tested for proper alignment, navigation, and responsiveness.
- Checked system compatibility across different browsers and devices.
- Ensured forms were user-friendly with proper validation and error messages.

8.6 Security Testing

- Passwords were verified to be stored securely using encryption.
- Forms were checked for SQL Injection and Cross-Site Scripting (XSS) prevention.
- CSRF token protection was verified in appointment booking and form submissions.

8.7 Results of Testing

- All core functionalities worked as expected.
- The system successfully handled appointment booking, doctor-patient interaction, and review submission without data loss.
- Minor issues like input validation and layout adjustments were fixed during testing.

9. Results & Discussion

- ❖ The medical appointment booking system has been successfully developed and tested. The integration of Laravel as the backend with a dynamic database made the system functional and reliable.

9.1 Results

- The system allows patients to register, log in, search doctors by specialization, and book appointments dynamically.
- Doctors can manage their profiles, update availability slots, and track patient reviews.
- The appointment process is smooth with role-based dashboards and secure authentication.

- Patients are able to provide ratings and reviews, which helps other users in decision-making.
- Security measures like password hashing and CSRF protection were successfully implemented.

9.2 Discussion

- The project achieved its main objective of converting a static medical website into a fully functional dynamic system.
- Compared to traditional manual appointment booking, this system saves time and improves patient-doctor communication.
- The role-based dashboard ensures that patients and doctors have access to only relevant features.
- Although most requirements were fulfilled, some advanced features such as payment integration, push notifications, and advanced analytics are still pending and can be considered in future improvements.

10. User Interface Screenshots

- ❖ The Medical Appointment Booking System has been designed with a simple and intuitive user interface. The layout ensures ease of navigation for both patients and doctors. The following are the main interfaces of the system:

10.1 Home Page

- **Hero Banner/Slider** – Highlights key features of the platform such as booking appointments, accessing qualified doctors, and categories of specialization. It creates a professional first impression.
- **Categories Section** – Displays various medical categories (e.g., cardiology, dermatology, pediatrics) so that patients can directly explore doctors by their specialization.
- **Call-to-Action Buttons** – Prominent buttons like “View All Doctors”, “Book Appointment Now”, and “Register/Login” guide users towards the next action.
- **About Us Section** – Briefly describes the purpose of the platform and how it helps patients and doctors.
- **Testimonials/Reviews Section** – Displays short reviews or feedback from patients to create credibility.

- **Footer** – Contains quick links, hospital/clinic details, contact information, and social media icons for extended connectivity.


[Home](#)
[Categories](#)
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Your Health, Our Priority

Connect with trusted medical professionals and book appointments with ease. Quality healthcare is just a click away.

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Medical Categories

Explore our comprehensive range of medical specialties



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Heart and cardiovascular system specialists
12 Doctors



Neurology
Brain and nervous system experts
8 Doctors



Ophthalmology
Eye care and vision specialists
6 Doctors



Dentistry
Oral health and dental care
15 Doctors

Meet Our Doctors

Experienced healthcare professionals dedicated to your well-being



Dr. Sarah Gupta
Cardiologist
15+ years experience
★★★★★ 4.9 (127 reviews)
[Book Appointment](#)



Dr. Michael Chen
Neurologist
12+ years experience
★★★★★ 4.8 (98 reviews)
[Book Appointment](#)



Dr. Emily Rodriguez
Pediatrician
10+ years experience
★★★★★ 4.9 (156 reviews)
[Book Appointment](#)

[View All Doctors](#)

About MeddyBuddy+

Revolutionizing healthcare accessibility through technology

Founded by a team of healthcare professionals and technology experts, we're committed to bridging the gap between patients and doctors through innovative digital solutions.

10,000+
Happy Patients

500+
Qualified Doctors

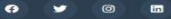
50+
Medical Specialties

24/7
Support Available





Your trusted partner in healthcare. Connecting patients with quality medical professionals for better health outcomes.



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Contact Info

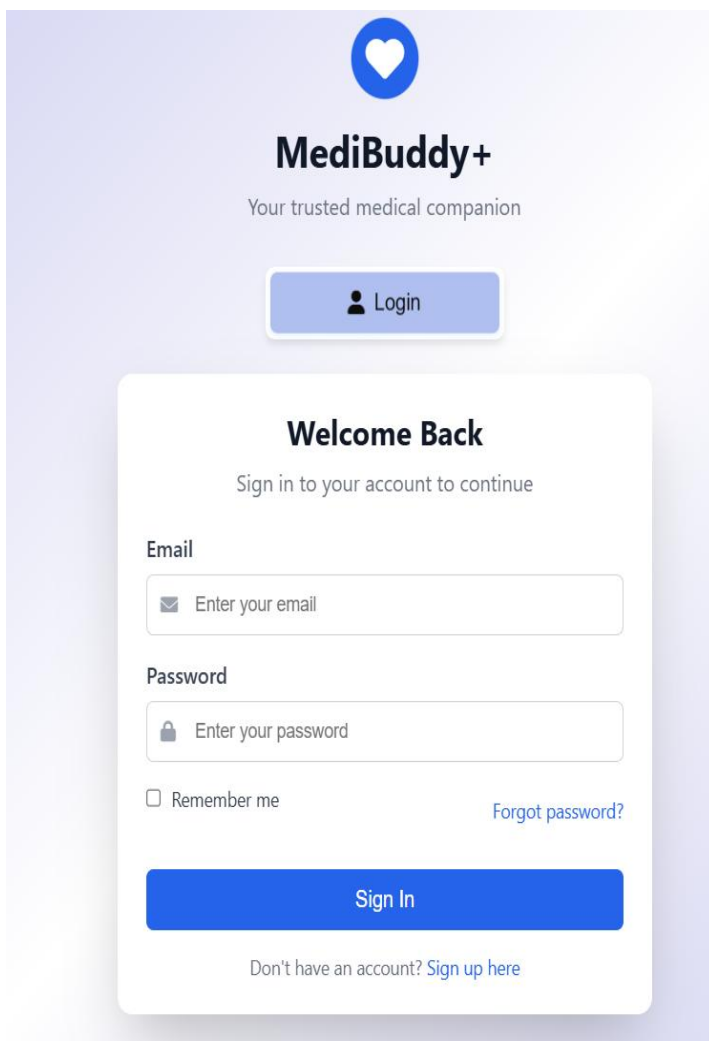
+91 6375874059

bhumikarathor.com

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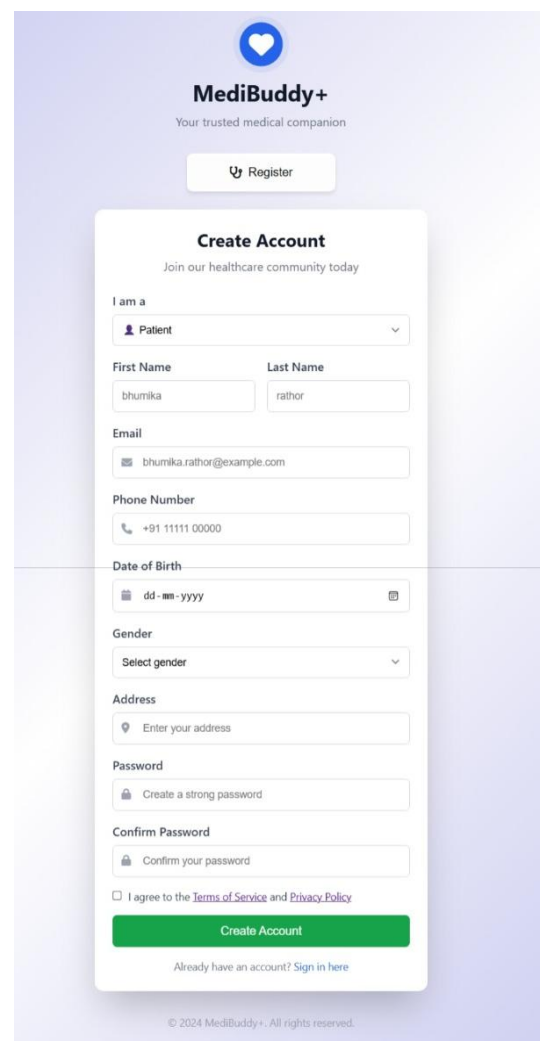
10.2 Registration and Login Page

- Provides common **registration forms for patients and doctors**.
- Requires user details such as name, email, password, and role selection.
- The login form validates credentials and redirects the user to the appropriate dashboard.
- **Role Selection Dropdown** – During registration, users must choose whether they are signing up as a **Patient** or **Doctor**, which ensures role-based access in the system.
- **Forgot Password Option** – Provides users with the ability to reset their password via email in case they forget their login credentials.
- **Error message show**- error messages show for invalid credentials, same email again used for register.



The login form for MediBuddy+ features a blue heart logo at the top. Below it, the text "MediBuddy+" and "Your trusted medical companion" are displayed. A blue "Login" button with a user icon is positioned above the main form. The form itself has a white background and a blue border. It contains fields for "Email" and "Password", both with placeholder text "Enter your email" and "Enter your password" respectively. A "Remember me" checkbox is located below the password field. A blue link "Forgot password?" is to the right of the password field. A large blue "Sign In" button is at the bottom. Below the button, a link "Don't have an account? Sign up here" is provided.

Login form



The registration form for MediBuddy+ features a blue heart logo at the top. Below it, the text "MediBuddy+" and "Your trusted medical companion" are displayed. A blue "Register" button with a user icon is positioned above the main form. The form itself has a white background and a blue border. It contains fields for "I am a" (a dropdown menu with "Patient" selected), "First Name" (with "bhumika" entered), "Last Name" (with "rathor" entered), "Email" (with "bhumika.rathor@example.com" entered), "Phone Number" (with "+91 11111 00000" entered), "Date of Birth" (with "dd-mm-yyyy" entered), "Gender" (a dropdown menu with "Select gender" selected), "Address" (with "Enter your address" entered), "Password" (with "Create a strong password" entered), and "Confirm Password" (with "Confirm your password" entered). A checkbox "I agree to the Terms of Service and Privacy Policy" is located below the password fields. A large green "Create Account" button is at the bottom. Below the button, a link "Already have an account? Sign in here" is provided.

Register form

10.3 Patient Dashboard

- Displays patient's **profile information, medical history, and appointment list.**
- Allows patients to **search doctors by specialization, book appointments, and manage bookings.**
- Patients can also **add reviews and ratings** for doctors after consultations.



Bhumika Rathor
 Patient ID: #MED-2024-001


Personal Information

Date of Birthjuly 12, 2008

Age17 years

GenderFemale

Phone(+91) 6375874059

Emailbhoomikarathor120708@gmail.com

Address123 kota rajasthan

Edit Information


Health Information

Blood TypeA+

Height5'7" (170 cm)

Weight125 lbs (57 kg)

Emergency Contact6375874059

Current Medications

Lisinopril 10mg

Metformin 500mg

Vitamin D3

Allergies

Penicillin

Shellfish


Appointments

Upcoming

History

Annual Physical Exam
 Dr. switta mittal • september 25, 2025 at 10:00 AM

Upcoming

Blood Work Follow-up
 Dr. devish patil • september 2, 2025 at 2:30 PM

Upcoming

Patient Dashboard

10.4 Doctor Dashboard

- Enables doctors to **manage their professional profile**, including specialization, qualifications, hospital name, consultation fees, and working hours.
- Doctors can **set schedules, view booked appointments, and update appointment status**.
- Reviews and ratings from patients are displayed for feedback.

Profile

[Edit Profile](#)

Dr. Sarah Johnson
Cardiologist

Qualifications: MBBS, MD (Cardiology) | Experience: 15 Years | Registration No.: MCI-12345

Languages: English, Spanish, French

Contact & Location [Edit](#)

Heart Care Medical Center
123 Medical Plaza, Suite 456
New York, NY 10001

Consultation Hours
Mon-Fri: 9:00 AM - 6:00 PM
Sat: 9:00 AM - 2:00 PM

+1 (555) 123-4567
Call for appointments

dr.johnson@medcare.com
Email for inquiries

[View on Google Maps](#)

About Dr. Sarah Johnson [Edit](#)

Dr. Sarah Johnson is a highly experienced cardiologist with over 15 years of practice in cardiovascular medicine. She specializes in interventional cardiology and has performed over 2,000 successful cardiac procedures. Dr. Johnson is committed to providing personalized care and staying at the forefront of cardiac treatment innovations.

Specialized Treatments

Angioplasty, Cardiac Catheterization, Pacemaker Implantation, Heart Valve Repair, Coronary Bypass, Arrhythmia Treatment

Areas of Expertise

Interventional Cardiology, Preventive Cardiology, Heart Failure Management, Cardiac Imaging

Awards & Memberships

- American College of Cardiology Fellow
- Best Cardiologist Award 2022
- Society for Cardiovascular Angiography Member
- International Heart Association Member

Appointment Settings

Consultation Fees
In-person: \$200 | Video call: \$150

Select Date
23 - 09 - 2025

Available Time Slots
9:00 AM, 10:00 AM, 11:00 AM, 2:00 PM, 3:00 PM, 4:00 PM

Consultation Mode
In-person, Video Call, Phone Call

[Edit Settings](#)

★ Patient Reviews

John Smith ★★★★★
"Excellent doctor! Very thorough examination and clear explanation of my condition. Highly recommended."

Maria Garcia ★★★★★
"Dr. Johnson saved my life with her expertise in cardiac surgery. Professional and caring approach."

Robert Wilson ★★★★★
"Great experience overall. The doctor was punctual and the treatment was effective."

Connect & Share

Follow Dr. Johnson

[Share Profile](#)

Doctor Dashboard

10.5 Doctor Detail Page

- Displays doctor's **detailed profile** including specialization, experience, consultation fees, hospital details, and contact number.
- Includes **ratings and reviews** given by patients.
- Shows **location map** and hospital/clinic address for easy navigation.



Dr. yaminee rathor

Cardiologist

Qualifications

MBBS, MD (Cardiology)

Experience

15 Years

Registration No.

MCI-12345

Languages

English Hindi

Contact & Location



Heart Care Medical Center
 123 Medical Plaza, Suite 456
 New York, NY 10001



Consultation Hours
 Mon-Fri: 9:00 AM - 6:00 PM
 Sat: 9:00 AM - 2:00 PM



91+ 6375874059
 Call for appointments



bhoomikarathor@gmail.com
 Email for inquiries

[View on Google Maps](#)

About Dr. yaminee rathor

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Specialized Treatments

Angioplasty

Cardiac Catheterization

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Preventive Cardiology

Heart Failure Management

Cardiac Imaging

Awards & Memberships

- American College of Cardiology Fellow
- Best Cardiologist Award 2022
- Society for Cardiovascular Angiography Member
- International Heart Association Member

Book Appointment

Consultation Fees

In-person: **200** Video call: **150**

Select Date

23 - 09 - 2025

Available Time Slots

9:00 AM

10:00 AM

11:00 AM

2:00 PM

3:00 PM

4:00 PM

Consultation Mode

In-person

Video Call

Phone Call

[Book Appointment](#)

★ Patient Reviews

John Smith

★★★★★

"Excellent doctor! Very thorough examination and clear explanation of my condition. Highly recommended."

Maria Garcia

★★★★★

"Dr. Johnson saved my life with her expertise in cardiac surgery. Professional and caring approach."

Robert Wilson

★★★★★

"Great experience overall. The doctor was punctual and the treatment was effective."

Connect & Share

Follow Dr. yaminee rathor






[Share Profile](#)

Doctor detail page

11. Security & Data Privacy Considerations

Since the system deals with sensitive medical and personal data of patients and doctors, ensuring security and privacy is a top priority. Various mechanisms have been implemented to protect data from unauthorized access and misuse.

11.1 Authentication and Authorization

- Users are required to register and log in with valid credentials.
- **Role-based access control** ensures that patients and doctors only access their respective dashboards.
- Unauthorized users cannot book appointments or view confidential details.

11.2 Password Security

- Passwords are never stored in plain text; instead, they are **encrypted using hashing algorithms** (such as bcrypt in Laravel).
- Strong password validation rules are applied during registration.

11.3 Data Encryption

- Sensitive information transmitted between client and server is secured using **HTTPS protocol**.
- Data at rest in the database is protected with secure configurations.

11.4 Protection Against Attacks

- **CSRF tokens** are implemented in all forms to prevent cross-site request forgery.
- Input validation and query binding protect the system from **SQL injection**.

11.5 Patient Data Privacy

- Medical history, personal details, and consultation records of patients are only visible to the respective user and doctor involved.
- No patient data is shared with third parties without consent.

11.6 Regular Backups and Recovery

- Database backup policies ensure that patient and doctor information is not lost due to system failure.
- Recovery mechanisms allow restoration of data when needed.

12. Team Contribution / Internship Learnings

12.1 Team Contribution

- ❖ The project was developed as part of an internship, where each member of the team contributed towards different aspects of the system. The division of responsibilities ensured smooth progress and better collaboration.
 - **Frontend Development:** Designing the user interface using HTML, CSS, and JavaScript. This included creating responsive layouts, navigation menus, forms, and dashboards.
 - **Backend Development:** Setting up Laravel framework, configuring routes, controllers, and middleware for smooth data handling.
 - **Database Management:** Designing and implementing relational tables such as users, doctor profiles, patient profiles, appointments, reviews, and schedules.
 - **Testing & Debugging:** Conducting unit testing, integration testing, and role-based testing to ensure the system works as expected.
 - **Documentation:** Preparing reports, flow diagrams, and user manuals to present the project in a structured format.

Each member collaborated and communicated regularly, which helped in identifying problems early and solving them efficiently.

12.2 Internship Learnings

- ❖ During the internship, I gained practical knowledge of web development and software project execution. Some of the key learnings are:
 - Learned the basics of **HTML, CSS, and JavaScript** for designing the frontend of a website.
 - Gained hands-on experience in **PHP** programming for server-side development.
 - Explored the **Laravel framework** and understood how to build dynamic web applications.
 - Learned how to design and manage a **MySQL database** with multiple relational tables.
 - Understood the concept of **role-based dashboards** for patients and doctors.
 - Implemented **basic security features** like password hashing and CSRF protection.

- Improved **problem-solving skills** and gained exposure to teamwork in real projects.

13. Conclusion

The Medical Appointment Booking System was successfully developed to convert a static medical website into a fully dynamic and functional platform. The system allows patients to register, log in, search doctors, and book appointments, while doctors can manage their profiles, set availability schedules, and view patient reviews.

The integration of **HTML, CSS, and JavaScript** for the frontend with **PHP and Laravel** for the backend ensured smooth functionality and responsiveness. The use of a **MySQL relational database** helped in maintaining structured data for users, doctors, appointments, and reviews.

The system not only fulfilled the main objective of simplifying the appointment booking process but also ensured **security, privacy, and role-based access** for different users. The dashboards provided to patients and doctors enhanced usability and improved the overall interaction between them.

Although the project achieved its core goals, there is still scope for future improvements, such as integrating payment gateways, sending appointment reminders, and developing a mobile application for wider accessibility.

In conclusion, the project provided valuable learning in terms of **web development, database management, and security practices**. It can be extended further into a fully professional medical management platform.

14. Future Scope

- ❖ The Medical Appointment Booking System has been designed as a fully functional platform; however, there are several areas where it can be extended and improved in the future. Some of the possible enhancements include:

1. Payment Gateway Integration

- Allow patients to pay consultation fees online through UPI, credit/debit cards, or net banking.

- Secure and hassle-free transactions will make the system more professional.

2. **Automated Notifications**

- SMS and email reminders for upcoming appointments.
- Notifications for appointment confirmation, cancellation, or rescheduling.

3. **Mobile Application Development**

- A dedicated **Android/iOS app** can be built for patients and doctors.
- Improves accessibility and convenience for users on the go.

4. **Telemedicine Features**

- Integration of **video calling and chat** features for online consultations.
- Especially useful for remote patients who cannot visit in person.

5. **Advanced Analytics and Reports**

- Doctors can get detailed reports on patient visits, revenue, and feedback.
- Patients can track their medical history and consultation trends.

6. **Multi-language Support**

- Adding different languages for patients from diverse regions.
- Makes the platform more user-friendly and widely acceptable.

7. **Integration with Hospital Management Systems (HMS)**

- Connect the appointment system with existing hospital software.
- Enables seamless handling of patient records, billing, and treatment history.

15. References

1. Laravel Framework Documentation - <https://laravel.com/docs>
2. PHP Official Documentation- <https://www.php.net/docs.ph>
3. MySQL Documentation - <https://dev.mysql.com/doc>
4. W3Schools – HTML, CSS, JavaScript Tutorials- <https://www.w3schools.com>
5. MDN Web Docs – JavaScript and Web APIs- <https://developer.mozilla.org>
6. GeeksforGeeks – Web Development Tutorials- <https://www.geeksforgeeks.org>
7. Articles, blogs, and online resources related to **medical appointment systems and telemedicine platforms**.

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